

VISWA M

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PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on experience in mechanical product design, CAD development, engineering analysis, simulation, testing support, and manufacturing workflows. Currently working as a Graduate Engineer Trainee in the Worm Gearbox Design team at Shanthi Gears Limited. Skilled in product development, prototype support, technical documentation, engineering calculations, testing methodologies, and cross-functional collaboration. Proficient in SolidWorks, Siemens NX, Creo, ANSYS, ERP/PLM systems, and engineering documentation with strong interest in innovative and cost-effective product engineering solutions.

EDUCATION

Sri Krishna College of Technology, Coimbatore Bachelor of Engineering in Mechanical Engineering	2022 – 2026 CGPA: 7.62/10
Alagar Public School (CBSE), Tuticorin Class XII	2021 – 2022 Percentage: 60.6%
Amrita Vidyalayam (CBSE), Ramnad Class X	2019 – 2020 Percentage: 77.6%

TECHNICAL SKILLS

- **CAD & Product Design:** SolidWorks, Siemens NX CAD, Creo Parametric, AutoCAD 2D, Assembly Modeling, Technical Drawings, GD&T
- **Engineering Analysis:** ANSYS Workbench, ANSYS APDL, Finite Element Analysis (FEA), Structural Analysis, Engineering Calculations
- **Testing & Validation:** Prototype Support, Simulation Fundamentals, Design Validation, Technical Reviews, Product Testing
- **Mechanical Engineering:** Product Development, Worm Gearbox Design, Manufacturing Processes, Tooling Fundamentals, Process Optimization
- **Documentation & Systems:** Oracle ERP, SLGPDM, PLM Systems, BOM Management, Engineering Documentation, Technical Reporting
- **Programming Languages:** Python, C++, Basic Java
- **Professional Skills:** Problem Solving, Cross-functional Collaboration, Risk Evaluation, Continuous Improvement, Technical Communication

PROFESSIONAL EXPERIENCE

Graduate Engineer Trainee – Design Shanthi Gears Limited (Murugappa Group), Coimbatore	Apr 2026 – Present
• Assisted in worm gearbox product design and development activities including CAD modeling, assembly validation, and engineering drawing preparation. • Supported product lifecycle activities through BOM preparation, engineering documentation, and design revisions using Oracle ERP and SLGPDM systems. • Participated in engineering analysis, simulation support, and technical reviews to improve product quality and manufacturing feasibility. • Collaborated with manufacturing and quality teams for corrective actions, defect resolution, and process optimization activities. • Supported prototype evaluation, product testing, engineering change management, and technical documentation activities. • Assisted in maintaining engineering standards, drawing consistency, and reusable design practices across projects.	

INTERNSHIPS

Research and Development Intern Super Auto Forge Pvt. Ltd., Chennai	May 2025
• Assisted in forging stage design, die development, and CAD modeling using Siemens NX software.	

- Supported tooling design, product development, and manufacturing process optimization for cold and warm forging applications.
- Conducted defect prediction and material flow simulations using DEFORM and Simufact software to improve process efficiency.
- Participated in engineering analysis, technical reviews, and process validation activities for manufacturing improvement.
- Assisted in maintaining engineering documentation, design records, and technical reports related to tooling and production workflows.
- Collaborated with production and quality teams to identify defects and support corrective engineering actions.

Student Intern

Jun 2024 – Jul 2024

National Institute of Ocean Technology (NIOT), Chennai

- Trained in ANSYS Workbench for simulation and analysis of marine energy system components and thermal systems.
- Learned fundamentals of Ocean Thermal Energy Conversion (OTEC) and Low Temperature Thermal Desalination (LTTD) systems.
- Assisted in engineering studies related to renewable energy systems, thermal analysis, and desalination technologies.
- Supported simulation-based evaluation and performance analysis activities for ocean-based energy applications.
- Participated in technical discussions, engineering documentation, and research-oriented learning activities.
- Gained exposure to innovative renewable energy technologies, industrial engineering workflows, and research methodologies.

PROJECTS

Modified Solar Dryer with Concave Fins and PCM Storage

Jan 2025 – Mar 2025

- Designed and developed a solar dryer integrated with PCM storage and concave fins for enhanced thermal efficiency.
- Performed thermal analysis, testing, and performance evaluation to optimize drying efficiency.
- Improved airflow distribution and energy utilization using forced convection mechanisms.
- Prepared engineering documentation and technical reports related to system performance and testing.

Wear Resistant Thermal Coating Development for Pantograph Applications

Jun 2025 – Mar 2026

- Conducted literature review and engineering analysis on HVOF sprayed Stellite-6 coatings.
- Studied corrosion resistance, wear resistance, thermal stability, and microstructural behavior of coated materials.
- Analyzed engineering properties and prepared technical documentation for coating performance evaluation.

PATENTS

Multi-Purpose Knife (Design Patent Granted)

- Designed a compact kitchen tool integrating knife, grater, and chip cutter functionalities.
- Improved safety, usability, and hygiene through innovative mechanical design enhancements.

Modified Solar Dryer with Concave Fins and PCM Storage (Utility Patent Applied)

- Developed an enhanced solar drying system with improved thermal energy storage capability.

CERTIFICATIONS

- Certified SolidWorks Associate (CSWA) – Dassault Systèmes
- Advanced Machining Process – NPTEL
- Additive Manufacturing – University of Michigan (Coursera)
- Geometric Dimensioning and Tolerancing (GD&T) – Udemy
- Fundamentals of Materials Science – Shanghai Jiao Tong University (Coursera)
- Complete Course in Creo Parametric – Udemy

ORGANIZATIONS

ISHRAE Student Member

Jun 2025 – Present

- Organized technical workshops and engineering events related to HVAC&R technologies.
- Collaborated with industry professionals and peers for technical knowledge sharing and networking.
- Promoted sustainable and energy-efficient engineering practices through technical activities.